

Chapter 4: Sequences Reference Sheet

Sequence Definition

A **sequence** is an ordered list of numbers: $a_0, a_1, a_2, a_3, \dots$. We can refer to this entire sequence with the notation $(a_n)_{n \in \mathbb{N}}$ or $(a_n)_{n \geq 0}$, or just (a_n) if we are being sloppy. Each subscript number is called the **index** of the sequence element (also called the *term of the sequence*). A sequence is a function on the domain of natural numbers $\mathbb{N}$ with a_n being the image element of the natural number n .

Closed Formula

A **closed formula** for a sequence $(a_n)_{n \in \mathbb{N}}$ is a formula for a_n using a fixed finite number of operations on n .

Recursive Definition

A **recursive definition** for a sequence $(a_n)_{n \in \mathbb{N}}$ is an equation defining the value of an element of a sequence a_n in terms of previous terms in the sequence together with explicit values, called **initial conditions**, for one or more initial terms (whose number must be one less than the number of terms in the equation). This equation and initial conditions together define a **recurrence relation**.

Common Sequences

Sequence	Definition
1, 4, 9, 16, 25, ...	The square numbers , $(s_n)_{n \geq 1}$, defined by $s_n = n^2$.
1, 3, 6, 10, 15, 21, ...	The triangular numbers , $(T_n)_{n \geq 1}$, defined by $T_n = \frac{n(n+1)}{2}$.
1, 2, 4, 8, 16, 32, ...	The powers of 2 , $(a_n)_{n \geq 0}$, defined by $a_n = 2^n$.
1, 1, 2, 3, 5, 8, 13, ...	The Fibonacci numbers defined recursively by $F_n = F_{n-1} + F_{n-2}$ with $F_1 = F_2 = 1$.

Sequence of Partial Sums Definition

Given a sequence $(a_n)_{n \in \mathbb{N}}$ we define the **sequence of partial sums** of (a_n) , $(b_n)_{n \in \mathbb{N}}$ as:

$$b_n = a_0 + a_1 + a_2 + \dots + a_n = \sum_{i=0}^n a_i.$$

Sequence of Difference of a Sequence

Given a sequence $(a_n)_{n \in \mathbb{N}}$ we define the **sequence of difference** of (a_n) , $(d_n)_{n \in \mathbb{N}}$ as:

$$d_n = a_{n+1} - a_n,$$

giving us:

$$a_{n+1} = a_n + d_n.$$

Arithmetic Sequence

If the terms of a sequence differ by a constant, we say the sequence is **arithmetic**. If the initial term (a_0) of the sequence is a and the common difference is d , then we have,

Recursive Definition of an Arithmetic Sequence

$$a_n = a_{n-1} + d \text{ with } a_0 = a.$$

Closed Formula of an Arithmetic Sequence

$$a_n = a + dn.$$

Geometric Sequence

A sequence is called geometric if the ratio between successive terms is constant. Suppose the initial term a_0 is a and the **common ratio** is r . Then we have,

Recursive Definition of a Geometric Sequence

$$a_n = ra_{n-1} \text{ with } a_0 = a.$$

Closed Formula of a Geometric Sequence

$$a_n = a \cdot r^n.$$

Characteristic Roots

Given a recurrence relation $a_n + \alpha a_{n-1} + \beta a_{n-2} = 0$, the **characteristic polynomial** is

$$x^2 + \alpha x + \beta$$

giving the **characteristic equation**:

$$x^2 + \alpha x + \beta = 0.$$

if r_1 and r_2 are two distinct roots of the characteristic polynomial (i.e., solutions to the characteristic equation), then the solution to the recurrence relation is

$$a_n = ar_1^n + br_2^n,$$

where a and b are constants determined by the initial conditions.

Characteristic Root Technique for Repeated Roots

Suppose the recurrence relation $a_n = \alpha a_{n-1} + \beta a_{n-2}$ has a characteristic polynomial with only one root r . Then the solution to the recurrence relation is

$$a_n = ar^n + bnr^n$$

where a and b are constants determined by the initial conditions.

Induction Proof Structure

To prove that a given statement $P(n)$ is true for all $n \geq 0$, you must prove two facts:

1. **Base case:** Prove that $P(0)$ is true. This can be proven directly and is often easy.
2. **Inductive case:** Prove that $P(k) \implies P(k + 1)$ for all $k \geq 0$. Assume $P(k)$ (the **inductive hypothesis**) and use that to prove $P(k + 1)$.

Assuming you can establish both of these parts, you conclude, "Therefore by the principle of mathematical induction, the statement $P(n)$ is true for all $n \geq 0$."